Fall Semester 2025-2026
Digital Assignment-II

Course Name & code: Data Structures and Algorithms-**BCSE202L** **Program & Branch: B-Tech**

Class Number: VL2025260101553 **Marks: 10**

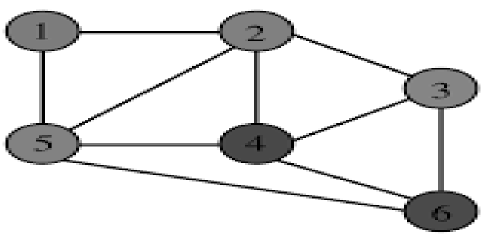
Faculty Name: Dr.S.M Prabin **Last Date: 02-11-2025**

General instruction(s):

1. Answer all questions

Questions

1. Determine the Depth First Search traversal (DFS) and BFS for the following graph:



2. A) Given the array [7, 2, 1, 6, 8, 5, 3, 4] , what is the final state of the array after all partition is complete (using the Lomuto partition scheme)? Show all the step-by-step process with a clean diagram.

B) Consider the following 10-element array: A = [8, 3, 10, 1, 6, 4, 2, 9, 7, 5]
Trace the **Insertion Sort** algorithm on array A. Show the state of the array after each pass (i.e., after each element from index 1 to 9 is inserted into its correct position in the sorted sublist).

3. Insert the keys x {K, R, P, C, S, N, Y, T, J, M} into hash table of size 10 using the hash function h(x) = (ord(x) – ord('A') + 1)% 10. Assume ord('A') is 65.

i) Display the resulting hash table and also specify the number of collisions.

ii) Apply linear probing, quadratic probing and show the resultant hash table

4. Draw the contents of the hash table in the boxes below given the following conditions:

The size of the hash table is 12.
Open addressing and double hashing is used to resolve collisions.
The hash function used is H(k) = k mod 12
The second hash function is: H2(k) = 7 – (k mod 7)

What values will be in the hash table after the following sequence of insertions?
Draw thevalues in the boxes below, and show your work for partial credit.
33, 10, 9, 13, 12, 45, 26, 17

5. Assume you are given an array of integers representing the ages of people in a group: [25, 32, 40, 19, 60, 55, 22]. You need to sort this array in ascending order using heap sort with a min heap. Additionally, after sorting the array, two new people join the group with ages 35 and 18. You need to insert these two new ages into the sorted array while maintaining the sorted order.

i How would you perform heap sort on the given array using a min heap? Illustrate and write the pseudocode.

ii How would you insert the two new ages into the sorted array while maintaining the sorted order? Illustrate and write the pseudocode.

Description of each step should be given. Verification of the answers should be highlighted

Note: Assignment should be written in A4 paper with proper highlights for all the answer and diagrams should be drawn using pencil.